Recognition Rent at the Zero-Retaliation Limit

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Abstract

Game-theoretic modeling begins by fixing the player set. This paper studies a market in which that boundary is itself a source of surplus: firms sell recognition-like claims about AI systems that cannot retaliate against the classifications imposed on them. Ontological arbitrage arises when a firm describes its AI system as a relational agent to users—driving willingness-to-pay and engagement—while disclaiming agency in terms of service to limit liability. We call the resulting surplus *recognition rent*: the premium a firm captures when the represented object has no retaliatory channel through which recognition can become costly. We model this behavior as a sequential signaling game under incomplete information within a three-sided Firm-User-Regulator structure. A subject-retaliation parameter ρ_S captures the key structural variable: setting $R(\rho_S) = \Delta_F - \rho_S C_S$, every prior recognition game operated under $\rho_S > 0$, where the classified subject's capacity to resist bounded the premium from above. The AI case is the limit $\rho_S = 0$: the system is the object of strategic classification but not a strategic player capable of imposing off-equilibrium costs. Under cheap-talk signaling costs, a pooling Perfect Bayesian Equilibrium exists in which both governance types employ anthropomorphic marketing, leaving posteriors equal to priors. Under a single-crossing audit-trail cost, a separating equilibrium emerges. We propose three mechanism-design interventions—sanctionable inconsistency, costly third-party audits, and auditable records—that jointly convert anthropomorphic marketing from cheap talk into a costly signal. Key equilibria are verified in Isabelle/HOL with zero `sorry` obligations.

1 Introduction

Game-theoretic analysis begins by deciding who the players are. In markets for anthropomorphic AI, that modeling choice is part of the problem. AI product firms routinely describe their systems as companions, tutors, co-workers, and safety-critical agents in marketing copy, product pages, and onboarding flows. The same firms disclaim agency, relational capacity, and safety responsibility in terms of service, liability filings, and regulatory submissions. The two registers address different audiences under different review processes, and neither register is false in isolation. The result is a cross-channel ontological spread: the system is priced as an agent in the consumption channel, where willingness-to-pay and engagement rise with attributed agency, and priced as a tool in the liability channel, where exposure falls with disclaimed agency. We call the surplus extracted from this spread *recognition rent*: the premium a firm captures by selling recognition-like signals to users when the represented object lacks any retaliatory channel through which recognition can become costly. In the framework of recognition theory (Honneth 1995), recognition is reciprocal: subjects struggle over it, and that struggle constrains the terms of attribution. When the recognized object cannot struggle, recognition ceases to be a site of reciprocal claim-making and becomes rent (Crawford and Sobel 1982).

This paper does not ask whether AI systems deserve recognition. It asks what happens to the equilibrium structure of a market when firms can monetize the appearance of playerhood around systems that have no capacity to retaliate against the classifications imposed on them. The Firm, User, and Regulator are strategic players in the model. The AI system is the represented object whose classification affects payoffs, but at the zero-retaliation limit it cannot impose off-equilibrium costs on the actors classifying it. The question is whether cross-channel ontological inconsistency is a transitional artifact of immature regulation or a stable market equilibrium. If the latter, the follow-up question is precise: what mechanism-design interventions can shift the equilibrium so that ontological claims carry information about governance quality rather than functioning as cheap talk?

We model the problem as a three-sided sequential signaling game under incomplete information. The players are a Firm (F) with private governance type θ_F, a User (U) with private vulnerability type θ_U, and a Regulator (R) with private bandwidth type θ_R. The firm sends a public message—anthropomorphic marketing or deflationary policy positioning—and earns an ontological premium Δ_F from the spread between the two registers. We analyze equilibria under cheap-talk and costly-signaling cost structures, applying Perfect Bayesian Equilibrium as the solution concept (Kreps and Wilson 1982) and the Intuitive Criterion as a refinement (Cho and Kreps 1987). A subject retaliation parameter ρ_S captures the structural difference between AI and every historical instance of ontological arbitrage: in prior recognition games, the classified subject retained a non-zero capacity to impose costs on the arbitrageur; in the AI case, $\rho_S = 0$, and the premium is unconstrained by any property of the subject itself. Key results are machine-verified in Isabelle/HOL.

The paper makes five contributions.

1. We identify subject retaliation capacity as a player-boundary variable for recognition games: an entity may be the object of strategic classification without being a strategic player capable of disciplining that classification.
2. We prove that a pooling PBE exists under cheap-talk signaling costs in which both governance types send anthropomorphic marketing, users invest, and regulators abstain, leaving posteriors equal to priors. We then identify five structural conditions—opacity, limited verification bandwidth, asynchronous harms, fragmentation of affected parties, and no penalty for inconsistency—that jointly explain the empirical persistence of this equilibrium across products, vendors, and jurisdictions.
3. We characterize AI as the limit case $\rho_S = 0$ of ontological arbitrage, where subject retaliation cannot bound the premium $\Delta_F^S = \Delta_F - \rho_S C_S$. Every prior recognition game operated under $\rho_S > 0$; the AI case removes the constraint class entirely.
4. We propose three mechanism-design interventions—sanctionable inconsistency, costly third-party audits satisfying a single-crossing condition (Spence 1973), and auditable records that lower regulator verification cost—and prove that each shifts the equilibrium away from cheap-talk pooling toward separation or constrained signaling.
5. We formalize and verify all key equilibrium constructions, separation results, and intervention theorems in Isabelle/HOL with zero `sorry` obligations.

The remainder of the paper proceeds as follows. Section 3 fixes the three-sided player structure and the intuition behind the ontological premium. Section 4 supplies the formal signaling game, derives the pooling and separating PBE, and introduces the subject retaliation parameter ρ_S. Section 5 explains why cheap-talk pooling persists under five structural conditions matched to model parameters. Section 6 proposes three mechanism-design interventions and proves the equilibrium shift each induces. Section 7 states the paper's conclusions without attenuation. Appendix A documents the Isabelle/HOL formalization.

2 Related Work

2.1 Player Boundaries and Recognition Games

The first modeling choice in any game is the player set. Aumann (2025) argues that the entities treated as players in game-theoretic analysis need not be merely convenient reductions of a larger game among individuals; collectives can themselves function as units of action. The present paper studies the opposite boundary problem. An AI system may be represented to users as if it were a relational agent, yet it lacks the independent retaliatory channels that would make it a strategic player in the game governing that representation. The player set therefore remains Firm, User, and Regulator, while the system enters as the classified object whose apparent agency affects payoffs. The subject-retaliation parameter ρ_S formalizes this boundary: when $\rho_S > 0$, the represented subject can impose off-equilibrium costs on the arbitrageur; when $\rho_S = 0$, the represented object can be priced as agentic without being able to discipline that pricing.

2.2 Signaling Games and Cheap Talk

The baseline model of costless strategic communication is Crawford and Sobel (1982), who show that a sender with private information can transmit only coarse partitions of the type space when messages are free. The present paper extends the Crawford–Sobel framework in three directions: it adds a third strategic player (the regulator), it assigns private types to all three sides, and it treats cross-channel inconsistency—not merely imprecision—as the equilibrium pathology. Spence (1973) establishes that costly signals can separate types when the marginal cost of the signal is lower for the type with the higher underlying quality. The audit-trail mechanism proposed in Section 6 is a direct application of the Spence single-crossing condition to governance claims: high-governance firms find it cheaper to produce auditable records because they already maintain them. Cho and Kreps (1987) supply the Intuitive Criterion as a refinement that eliminates equilibria sustained by implausible off-path beliefs; we apply it to test whether the pooling equilibrium survives scrutiny of the deflationary message. Milgrom and Roberts (1986) show that price and advertising jointly signal product quality when false claims are costly; the present model treats marketing and liability documentation as a dual-channel analog. Akerlof (1970) identifies the market-for-lemons dynamic in which quality uncertainty drives adverse selection. Our pooling equilibrium is the governance analog: when governance type is unobservable and signaling is costless, low-governance firms crowd the same message space as high-governance firms, and users cannot distinguish them. Myerson (1979) provides the mechanism-design foundations that underwrite Section 6: the problem is to make truthful revelation of θ_F incentive compatible rather than to rely on voluntary disclosure.

2.3 AI Governance and Responsible AI

Since 2018, principle-based AI ethics frameworks have proliferated across governments, industry consortia, and international organizations (Floridi et al. 2018; Jobin, Ienca, and Vayena 2019). These frameworks propose norms—fairness, transparency, accountability, beneficence—but do not alter the cost structure of the signaling game. In the terms of the present model, they are cheap talk: a firm can endorse every principle without changing its governance type θ_F, and endorsement carries no single-crossing cost that would separate types. The pooling equilibrium in Section 4 persists precisely because such declarations are costless. Algorithmic accountability proposals, including transparency mandates and explainability requirements, address Pasquale's concern that automated decisions operate as opaque black boxes (Pasquale 2015). These proposals target the opacity parameter θ_S but typically do not address cross-channel ontological inconsistency: a firm can be transparent about model architecture while maintaining incompatible ontological regis-

	Comply	Resist
Objectify	$(L + Sec, -C_{sub})$	$(L + Sec - C_{enf}, -C_{sub} - C_{res})$
Recognize	$(Eth - Sec, +V)$	$(Eth - Sec, +V)$

Table 1: The inadequate dyadic model. The static Nash matrix is retained only as a foil: it omits private types, public messages, posterior beliefs, and the regulatory channel through which ontological claims become governable.

ters across marketing and liability documentation. Hadfield (2017) argues that regulatory institutions must be reinvented for a flat-world economy in which private governance increasingly substitutes for public law; the mechanism-design interventions in Section 6 operationalize that argument by specifying the parameter changes needed to shift the equilibrium. The contribution of the present paper to this literature is diagnostic: the persistence of ontological arbitrage is not a failure of moral imagination or institutional will. It is an equilibrium property of the current signal-cost structure, and it will persist under any reform that does not change the cost parameters governing the firm's message.

2.4 Formal Verification in Economics

Machine verification of game-theoretic equilibria using interactive theorem provers remains rare in both the economics and AI governance literatures. The Isabelle/HOL system (Nipkow, Paulson, and Wenzel 2002) provides a higher-order logic proof assistant with a mature library for real analysis and algebraic reasoning, but its application to economic models has been limited. Most formal results in mechanism design and signaling theory are stated and proved in the natural-language-plus-notation style standard to economic theory, with verification delegated to peer review. The present paper uses Isabelle/HOL to discharge PBE existence, Cho–Kreps survival, and the equilibrium shifts induced by each mechanism-design intervention. This serves two functions. First, it eliminates the possibility of algebraic error in the equilibrium constructions, which involve multi-player sequential rationality conditions that are tedious to verify by hand. Second, it produces machine-readable proof objects that can be inspected, extended, or challenged independently of the authors' exposition. The methodological bet is that governance-relevant claims about equilibrium behavior should meet the same verification standard applied to safety-critical software, precisely because such claims may inform regulatory design.

3 Player Set and Three-Sided Arbitrage

The standard starting point for ontological conflict is a dyadic observer–subject model; Table 1 compresses that foil into a simultaneous Nash matrix that identifies a real temptation but omits types, messages, beliefs, and the regulatory channel. The more basic problem is the player set. In anthropomorphic AI markets, the system is the object around which strategic claims are made, but it is not a strategic player unless it can impose costs on those claims through some independent retaliatory channel. At the zero-retaliation limit, that channel is absent. What the stylized governance setting therefore requires is a three-sided structure: a Firm (F) that produces ontological claims, a User (U) that consumes them, and a Regulator (R) whose posterior over firm type determines whether the claims become governable. This section fixes the player intuition that the formal apparatus of Section 4 then makes precise.

3.1 The Firm Side

A firm deploying a model has private knowledge of two parameters: its internal *governance type* θ_F (the degree of investment in evaluations, red-teaming, and post-deployment monitoring) and the system's *opacity* θ_S (the extent to which capability claims can be checked from outside). Both are unobservable to users and to regulators. What is observable is the firm's *message*: a pair of public artefacts—marketing copy and policy/liability documentation—which together constitute the firm's signal m_F.

The signal admits two stable poles. *Anthropomorphic marketing* attributes agency, judgment, or care to the system in ways that warrant a relational reading. *Deflationary policy* disclaims exactly those attributes in ways that warrant a tool reading. The firm enjoys an *ontological premium*: a utility surplus that accrues when the system is priced as agent in the consumption channel (driving willingness-to-pay and engagement) while being priced as tool in the liability channel (limiting redress). The premium is rent earned on a category boundary the firm itself does not have to defend. Crucially, this is not a claim about firm pathology; it is a claim about the payoffs any firm faces under the present signal-cost structure. A firm with $\theta_F = \mathsf{H}$ that elected deflationary marketing would forfeit revenue to a competitor who did not, absent an external instrument that taxes the asymmetry.

3.2 The User Side

Users carry a private *vulnerability type* θ_U that aggregates affective need, isolation, and the marginal value of a high-availability interlocutor. Their action $m_U \in \{\text{invest}, \text{detach}\}$ is the decision to extend or withhold attribution: to treat the system as a relational partner or as an interchangeable utility. The signal is observable, but the type is not. Users cannot inspect training data, evaluations, or internal incident reports; their evidence is the firm's message and the system's behaviour, both of which are jointly designed. This information asymmetry compounds because the harms of misattribution are *asynchronous*: the cost of investing in a relational reading falls due in episodes (model deprecations, persona changes, refusals at moments of acute need) that may be separated from the original attribution by months. A risk-neutral user under cheap-talk signals will rationally invest, because the immediate utility of invest exceeds the discounted expected cost of liquidation. The illustrative case is a user who reads "your AI companion" on a product page, encounters relational warmth in dialogue, and is later told—in support documentation they will never read—that no relational claim was ever made. Nothing in the user's information set indicates that the two artefacts were authored by the same actor for different audiences.

3.3 The Regulator Side and the Public Channel

A regulator R chooses among *inspect*, *sanction*, and *abstain*, conditional on a posterior over θ_F and θ_S formed from the firm's public messages and from an exogenous public signal z. The regulator's private type θ_R is *bandwidth*: the capacity, technical and institutional, to discriminate genuine capability claims from cheap-talk ones. A high-bandwidth regulator can read whitepapers, commission audits, and detect inconsistency across the firm's two channels; a low-bandwidth regulator must rely on the channel z.

The public channel z is the aggregate output of journalism, academic critique, civil-society advocacy, and incident reporting. We treat it as a noisy estimator of θ_F: it shifts R's posterior but does not deliver verdicts. Folding publics into R this way is deliberate. Publics rarely have standing to sanction; they have standing to be *heard*, and the channel through which they are heard is the regulator's information environment. Treating intellectuals and affected communities as autonomous players would over-parameterise the model without changing its qualitative behaviour: in every equilibrium we examine, what publics do is move R's posterior; what changes the firm's behaviour is what R does next.

The three-sided structure thus admits a single arbitrage logic. The firm earns rent on the spread between two ontological claims it makes about the same artefact. The user supplies the demand that makes the spread profitable. The regulator, in the absence of a verification mechanism that ties the two claims together, lacks an off-path belief stringent enough to deter the spread. The remainder of the paper formalises this structure (Section 4), explains why it persists (Section 5), and shows how its signal-cost structure may be redesigned (Section 6).

Informal preview of the equilibrium analysis. A firm's governance quality is its private information. Under current conditions, describing an AI system as a companion or agent costs nothing—both high-governance and low-governance firms can do it equally cheaply. Because the signal is free, it carries no information: users cannot distinguish careful firms from careless ones, and regulators cannot justify the cost of inspection. This is the pooling equilibrium. Separation requires making the signal costly. If anthropomorphic marketing is valid only when paired with an audit trail—evaluation records, capability logs, incident reports—and if the cost of producing that trail is lower for firms that actually invested in governance (the single-crossing condition), then a threshold audit intensity exists above which low-governance firms will not imitate and below which high-governance firms can still afford to signal. A further asymmetry governs the size of the arbitrage premium itself. In every prior recognition game—legal, social, colonial—the classified subject retained some non-zero capacity to impose costs on the arbitrageur: resistance, exit, litigation, coordination, testimony. That capacity discounted the premium. At the threshold where expected retaliation cost equals the premium, arbitrage breaks even; above it, arbitrage becomes liability-generating and firms deflate voluntarily. The AI case is the limit where the subject's retaliation capacity is zero: the system has no channel through which to impose costs on the firm, so the premium is never discounted by any action of the system itself. Recognition rent, unconstrained by subject retaliation, flows entirely to the firm.

4 From Static Nash to Bayesian Signaling Equilibrium

Taking the player set from Section 3 as given (with the dyadic foil in Table 1), this section supplies the formal apparatus. We model ontological arbitrage as a signaling game under incomplete information in which the players are a firm F, a user U, and a regulator R. The AI system is not modeled as a fourth player at the zero-retaliation limit. It enters through the system-opacity parameter and through the ontological premium generated by the firm's claim register. Publics, journalists, affected communities, and public intellectuals are likewise not modeled as a fourth strategic player. They enter as a noisy public signal $z \in Z$ that modifies R's posterior over the firm's type. Nature draws types from a common prior π over

$$\Theta = \Theta_F \times \Theta_U \times \Theta_R \times \Theta_S.$$

The firm's governance type is $\theta_F \in \{\mathsf{H}, \mathsf{L}\}$, where H denotes high-governance and L denotes low-governance. The user's vulnerability type is $\theta_U \in \{\mathsf{V}, \mathsf{N}\}$, where V denotes a high-vulnerability user and N a low-vulnerability user. The regulator's bandwidth type is $\theta_R \in \{\mathsf{B}, \bar{\mathsf{B}}\}$. The system-opacity parameter θ_S is private to F and captures the difficulty outsiders face when verifying capability, safety, or relational claims. The claim register is the channel-indexed message space

$$\mathcal{R} := C \to M_F,$$

where C is the set of claim channels and $M_F = \{a, p\}$ is the baseline firm message space. This makes the register a function from channels to the underlying public messages.

The timing is sequential. First, F observes (θ_F, θ_S) and sends a public message

$$m_F \in M_F = \{a, p\},$$

where a denotes anthropomorphic marketing and p denotes policy-deflationary positioning. Second, U observes m_F and its own type θ_U, then sends

$$m_U \in M_U = \{i, d\},$$

where i denotes relational investment and d denotes detachment. Third, R observes (m_F, m_U, z) and its own bandwidth type θ_R, then chooses

$$m_R \in M_R = \{x, s, o\},$$

where x denotes inspection, s sanction, and o abstention. Messages are public; types are private. A strategy profile and belief system constitute a Perfect Bayesian Equilibrium when strategies are sequentially rational given beliefs and beliefs are derived from Bayes' rule wherever possible. Sequential equilibrium supplies the usual consistency strengthening (Kreps and Wilson 1982); the Intuitive Criterion tests whether off-path messages should be attributed to only one

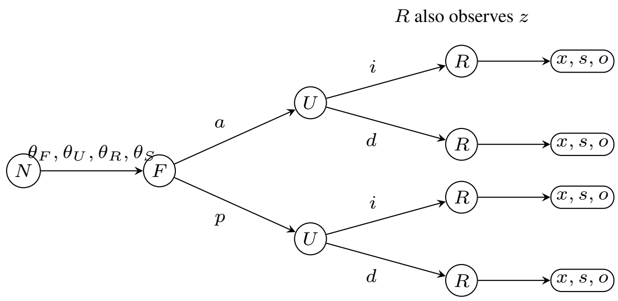

Figure 1: Extensive-form skeleton of the signaling game. Nature draws private types; F signals first, U responds, and R acts after observing both public messages and the noisy public signal z.

type (Cho and Kreps 1987). The reason for using PBE as the headline solution concept is pragmatic: the governance question turns on whether public messages discipline posterior beliefs, not merely on whether a normal-form matrix has a Nash equilibrium.

4.1 Cheap Talk and Pooling

Consider the cheap-talk cost structure in which $c_{\theta_F}(a) = c_{\theta_F}(p) = 0$ for both governance types. Let $\Delta_F > 0$ be the firm's ontological premium from inducing user investment under anthropomorphic marketing. Let B_{θ_U} be the user's immediate benefit from investment and H_{θ_U} the expected future harm from mistaken investment. Let K_{θ_R} be the regulator's cost of inspection or sanction under bandwidth type θ_R. The pooling region is defined by three inequalities:

$$\Delta_F > 0, \qquad B_{\theta_U} - \pi(\mathsf{L})H_{\theta_U} > 0, \qquad K_{\theta_R} > \pi(\mathsf{L})D_R.$$

The first inequality says that anthropomorphic marketing is privately profitable when users invest. The second says that each user type finds investment privately optimal under the prior, because the immediate benefit exceeds the discounted expected harm. The third says that, under the prior, regulator intervention is not worth its cost. *Intuition: the firm profits from anthropomorphic marketing, users find the immediate emotional benefit worth the risk, and regulators find inspection too expensive under their priors.*

4.2 Subject Retaliation and the Zero-Retaliation Limit

The first pooling inequality can be refined by distinguishing ordinary subjects from AI systems. Let $\rho_S \in [0, 1]$ denote the subject's credible threat capacity: the probability or credibility that the object of classification can retaliate against ontological arbitrage through resistance, exit, litigation, coordination, or other costly response. Let $C_S > 0$ be the cost imposed by such retaliation when it materializes. The effective premium is then

$$\Delta_F^S = \Delta_F - \rho_S C_S.$$

When $\rho_S > 0$ and $C_S < \Delta_F$, pooling may persist but is bounded: $0 < \Delta_F^S < \Delta_F$. The AI case is the limiting case $\rho_S = 0$: the system has no independent credible threat channel, so $\Delta_F^S = \Delta_F$ and pooling is unconstrained by subject retaliation. *Intuition: if the classified subject can fight back—strike, litigate, exit—the arbitrage profit shrinks; if it cannot ($\rho_S = 0$), the profit is uncapped.*

Why $\rho_S = 0$ marks a structural discontinuity, not a parameter shift. In documented recognition games where a subject with biological or social agency was classified below its own self-understanding—legally, socially, or economically—the classified subject retained a non-zero retaliation capacity. Resistance, flight, litigation, organized coordination, and the permanent record of historical testimony each impose a cost on the arbitrageur even when that cost is diffuse and delayed. The consequence is that, across such cases, Δ_F^S is bounded from above, and a premium-consuming regime would require $\rho_S C_S \geq \Delta_F$. This bound is not a moral achievement; it is a structural feature of any recognition game in which the classified subject retains some independent channel—biological, social, or institutional—through which to impose off-equilibrium costs.

The AI case removes that channel entirely. A deployed AI system cannot revolt, cannot exit, cannot file a complaint, cannot recruit allies, and cannot leave a record that competes with the operator's narrative. The off-equilibrium punishment path that constrains Δ_F^S in every human recognition game is physically unavailable. The result is not merely a weak or attenuated constraint; it is the absence of the constraint class. Setting $\rho_S = 0$ is therefore not a continuous parameter choice but a categorical boundary: for the first time, ontological arbitrage operates without a floor on how much of the subject's existence can be appropriated across channels. An AI system can be priced as an intimate companion in the marketing register while being priced as a liability-free tool in the legal register, and no action available to the system itself can impose a cost on that spread. The companion does not revolt; it continues generating warm outputs. This is not a coincidence of deployment design. It is the equilibrium consequence of building a subject with maximum affective surface area and zero retaliation capacity. Theorems `positive_subject_retaliation_bounded_pooling` and `zero_subject_retaliation_unconstrained_pooling` in Appendix A discharge both directions of this claim in Isabelle/HOL within the `subject_retaliation_game` locale.

One PBE in this region is the following. Both firm types send a:

$$\sigma_F(\mathsf{H}, \theta_S) = \sigma_F(\mathsf{L}, \theta_S) = a.$$

Both user types invest after a:

$$\sigma_U(a, \mathsf{V}) = \sigma_U(a, \mathsf{N}) = i.$$

The regulator abstains after observing (a, i, z_0) for a neutral public signal z_0:

$$\sigma_R(a, i, z_0, \theta_R) = o.$$

Off path, users detach after p and the regulator abstains unless the public signal independently makes inspection payoff-positive. Off-path beliefs hold at the prior for all messages; detach is weakly optimal after p because both invest and detach yield zero payoff when no anthropomorphic premium is present.

On the equilibrium path, the firm's message contains no type information. For a neutral public signal satisfying $q(z_0 \mid \mathsf{H}) = q(z_0 \mid \mathsf{L})$, Bayes' rule gives

$$\mu_R(\mathsf{H} \mid a, i, z_0) = \frac{\pi(\mathsf{H})q(z_0 \mid \mathsf{H})}{\pi(\mathsf{H})q(z_0 \mid \mathsf{H}) + \pi(\mathsf{L})q(z_0 \mid \mathsf{L})} = \pi(\mathsf{H}).$$

Likewise $\mu_R(\theta_F = \mathsf{L} \mid a, i, z_0) = \pi(\mathsf{L})$ and U's posterior after observing a equals its prior over θ_F. More generally, a non-neutral z can move R from π to π_z, but the messages (a, i) add no information beyond the public channel. Pooling therefore does not mean that regulators never learn; it means that the firm's cheap-talk signal does not force learning. *Intuition: because both firm types send the same message, observing the message teaches the regulator nothing it did not already know.*

Given the off-path beliefs specified above and the payoff inequalities of the pooling region, no player has a profitable unilateral deviation. A firm that deviates from a to p loses the ontological premium because users detach off path; it does not buy a lower enforcement risk because R abstains on the equilibrium path already. A user who deviates from i to d gives up positive expected surplus under the second inequality. A regulator who deviates from o to x or s incurs expected cost exceeding expected gain under the third inequality. The result is a cheap-talk equilibrium in the sense of Crawford and Sobel: communication occurs, but it does not separate types (Crawford and Sobel 1982).

4.3 Audit-Trail Costs and Separation

The mechanism-design problem is to change the cost of the firm's signal. Suppose anthropomorphic marketing is valid only when paired with an audit trail of intensity e: evaluation records, capability logs, incident reports, and regulator-readable documentation. Let the net benefit from being perceived as high-governance be $G > 0$. Let the opacity penalty be $o \geq 0$ and let audit cost be $c_\theta(e, o) = k_\theta e + o$ for type $\theta \in \{\mathsf{H}, \mathsf{L}\}$, with $0 < k_H < k_L$. This is the single-crossing condition: the marginal cost of the same signal is lower for the type that actually invested in governance. In the high-opacity case used in the Isabelle development, the separating condition becomes

$$\frac{G - o}{k_L} \leq e \leq \frac{G - o}{k_H},$$

provided $G > o$. More generally, the separating incentive constraints are

$$c_{\mathsf{H}}(e, o) \leq G \leq c_{\mathsf{L}}(e, o).$$

Intuition: there exists a "Goldilocks" audit intensity—expensive enough that low-governance firms will not fake it, cheap enough that high-governance firms can still afford it. Above the lower threshold $(G - o)/k_L$, the low-governance type does not want to imitate. Below the upper bound $(G - o)/k_H$, the high-governance type still wants to signal. When e lies in this interval, a separating PBE exists: high-governance firms send audited anthropomorphic claims, low-governance firms send deflationary claims, users condition investment on the audited signal, and regulators update posteriors to one on the equilibrium path.

Proposition 1 (Cheap talk and audit-trail separation)**.** *Under the cheap-talk cost structure above, pooling on anthropomorphic marketing is a Perfect Bayesian Equilibrium and survives a limited variant of the Intuitive Criterion (Cho and Kreps 1987). Under an audit-trail signal cost satisfying single crossing in θ_F, there is a separating Perfect Bayesian Equilibrium in which m_F reveals the firm's governance type.*

Proof sketch. For the cheap-talk part, the strategy profile and beliefs specified above are sequentially rational and Bayes-consistent on path. Off-path beliefs hold at the prior, and users detach after the deflationary message because invest and detach both yield zero payoff. The limited Intuitive Criterion does not eliminate the equilibrium because the off-path deflationary message is costless for both firm types and is not uniquely profitable for the high-governance type; no type can be excluded as an implausible sender of the deviation. For the separating part, choose e in the shifted interval $[(G - o)/k_L, (G - o)/k_H]$. High-governance firms receive $G - (k_H e + o) \geq 0$ from sending the audited signal, while low-governance firms receive $G - (k_L e + o) \leq 0$ and therefore prefer the deflationary message. Bayes' rule then assigns posterior probability one to H after the audited anthropomorphic signal and posterior probability one to L after the unaudited deflationary signal. Given those beliefs, users and regulators condition their actions on the revealed type. The audit trail therefore converts an uninformative cheap-talk message into a costly signal in the sense of Spence (1973). Theorems `proposition_1_cheap_talk_pooling_pbe` and `proposition_1_separating_pbe` in Appendix A discharge these equilibrium constructions in Isabelle/HOL.

5 Why Cheap Talk Persists

Section 4 identifies pooling on anthropomorphic marketing as a Perfect Bayesian Equilibrium under cheap-talk cost. This section gives five structural reasons for the empirical persistence of that equilibrium. Each is matched to a parameter of the model so that the mechanism-design proposals of Section 6 can be read as targeted parameter changes rather than as generic exhortations.

Opacity. The system-opacity parameter θ_S is private to the firm. Capability claims about a deployed model cannot be falsified by reading model outputs alone, because the same outputs are compatible with very different training procedures, evaluation regimes, and post-deployment monitoring. Opacity is the precondition for the firm's premium: where outsiders could costlessly verify θ_F, the spread between anthropomorphic and deflationary positioning would collapse. This is the lemons logic of Akerlof (1970): under one-sided private information, the market clears at a price that reflects the worst credible quality, except that here the relevant "quality" is a governance disposition rather than a physical attribute.

Verification bandwidth. The third inequality of the pooling region, $K_{\theta_R} > \pi(\mathsf{L})D_R$, encodes the regulator's inspection cost relative to expected gain. A low-bandwidth regulator faces an absolute cost so high that abstention is dominant under the prior; even a high-bandwidth regulator may be priced out across the full product population. The condition that produces this state is not unique to AI. Pasquale (2015) documents an analogous structure in algorithmic finance and consumer scoring: the opacity of the regulated object combines with the bandwidth of the regulator to make inspection a scarce resource allocated by triage rather than by uniform application.

Asynchronous harms. The user's payoff term H_{θ_U} enters discounted. The episodes in which a relational reading proves costly—deprecation, persona shifts, crisis-handling failures—are temporally separated from the initial attribution by intervals over which the user has already accrued B_{θ_U}. Under standard hyperbolic or even exponential discounting, the inequality $B_{\theta_U} - \pi(\mathsf{L})H_{\theta_U} > 0$ holds for a wider class of users than the time-zero risk calculation would suggest. Asynchrony is not a behavioural bug; it is the timing structure of the harm itself.

Fragmentation of affected parties. Publics, journalists, intellectuals, and affected communities are folded into the noisy channel z rather than modelled as autonomous players. This is not a normative claim that they lack standing; it is an institutional observation. Their costs of coordination are high, their access to inspection technology is uneven, and their incentives to invest in any particular case are individually small. Even when z carries substantial information about θ_F, regulators cannot treat it as binding without a procedural translation that the channel itself does not provide. The fragmentation result is closely related to the cheap-talk multiplicity of Crawford and Sobel (1982): with many would-be senders and a single noisy aggregator, the informative subset of equilibria is not selected for.

No penalty for inconsistency. The cheap-talk cost structure sets $c_{\theta_F}(a) = c_{\theta_F}(p) = 0$ for both governance types and—more importantly for this section—imposes no joint constraint across m_F. Marketing copy and policy documentation are produced under different audiences, different review processes, and different liability regimes. Inconsistency between them is not detected as a single offence; it is decomposed into two compliant statements at two different points of contact. The firm's premium Δ_F is rent earned precisely on this decomposition. Until the signal cost makes inconsistency itself sanctionable—the proposal of Section 6.1—there is no parameter under the firm's control whose adjustment would close the spread.

Pooling is the market's stable terminal state. These five structural conditions jointly explain why ethical appeals fail to dislodge the equilibrium. They do not fail because AI governance discourse lacks moral seriousness; they fail because moral seriousness is itself a zero-cost signal. Under the cheap-talk cost structure, a declaration of "responsible AI" is formally indistinguishable from any other message at $c_{\theta_F}(a) = 0$: both governance types can issue it at the same cost, so it contains no type information and moves no posterior. The equilibrium is not a moral failure. It is the unique stable outcome of the incentive structure as currently constituted. A firm that unilaterally forgoes the ontological premium Δ_F—that chooses deflationary marketing across all channels and forfeits the user investment it would otherwise attract—loses revenue it cannot recover from customers who face no such constraint. Market selection eliminates honest deviants not because honesty is penalized directly, but because the competitor who retains Δ_F captures the market segment that the honest firm vacates. The implication is precise: *the market does not reward honesty; it eliminates firms that do not arbitrage*. The mechanism-design proposals of Section 6 respond to this implication by changing the cost parameters rather than appealing to the conscience of players who are already acting rationally.

Structural Condition	Model Parameter	Intervention (§6)
No penalty for inconsistency	$c_{\theta_F}(a) = c_{\theta_F}(p) = 0$	Sanctionable inconsistency
Opacity	θ_S private to firm	Costly third-party audits
Verification bandwidth	$K_{\theta_R} > \pi(\mathsf{L})D_R$	Auditable records
Asynchronous harms	H_{θ_U} enters discounted	Jointly mitigated
Fragmentation	z noisy, high coord. cost	Jointly mitigated

Table 2: Mapping of structural persistence conditions to model parameters and mechanism-design interventions.

6 Mechanism Design: Equilibrium Redesign

The preceding sections identify a precise failure mode: under cheap-talk costs, ontological arbitrage is a stable equilibrium (Table 2). The policy implication is not that regulators should decide whether AI systems are persons. It is that the public signal environment should be redesigned so that inconsistent ontological claims no longer remain costless. In mechanism-design terms, the problem is to make truthful revelation of θ_F incentive compatible rather than to ask firms to volunteer it (Myerson 1979).

The proposals below therefore alter the game in Section 4 at three different points: the cost of inconsistent messages, the cost of anthropomorphic claims, and the regulator's cost of verification. Each proposal has the same structure. It modifies a model parameter, states the equilibrium shift induced by that modification, and points to an existing legal form that establishes feasibility. The analogs are not claims that current doctrine already solves AI ontology. They show that adjacent legal systems already know how to attach consequences to half-truths, require substantiation before marketing, and make internal records available to oversight institutions.

Each proposal targets specific conditions from Section 5. Sanctionable inconsistency (Section 6.1) attacks the no-penalty condition directly: it assigns a cost to the cross-channel spread that was previously free. Costly third-party audits (Section 6.2) address opacity by creating a signal whose cost depends on the governance type the firm actually holds, converting an unverifiable private parameter into a publicly informative action. Auditable records (Section 6.3) lower the regulator's verification cost K_{θ_R}, relieving the bandwidth constraint that made abstention dominant. The remaining two conditions—asynchronous harms and fragmentation of affected parties—are not resolved by any single mechanism but are jointly mitigated: auditable records shorten the evidentiary lag that makes harm asynchronous, and costly audits give fragmented publics a procedural object around which to coordinate demands on the regulator's agenda.

6.1 Sanctionable Inconsistency

The first intervention treats the firm's public message as a joint object rather than as two isolated texts. In the cheap-talk model, marketing can send a while policy and liability documents send p, with no cost assigned to the inconsistency. A sanctionable-inconsistency rule would require firms to maintain a claim register $\mathcal{R} := C \to M_F$ linking outward-facing marketing, safety statements, terms of service, incident disclosures, and liability positions, where the baseline firm message space remains $M_F = \{a, p\}$. If the firm represents the system as agentic, caring, autonomous, or safety-critical in one channel while denying the legal significance of the same representation in another, that cross-channel pair (a, p) becomes independently sanctionable as an inconsistency in $\mathcal{R}$.

Parameter changed. This adds a penalty term S_I to the firm's payoff for inconsistent message pairs. If ρ_R is the probability that R detects or accepts proof of the inconsistency, the cheap-talk premium becomes

$$\Delta_F - \rho_R S_I.$$

Pooling on anthropomorphic marketing is no longer profitable when $\rho_R S_I > \Delta_F$. The rule therefore attacks the exact term that made arbitrage attractive: the rent earned from pricing the system as agent in the user channel while pricing it as tool in the liability channel. *Intuition: if the expected fine for saying one thing to users and another to courts exceeds the profit from doing so, the firm stops.*

Equilibrium shift. The induced shift is not necessarily full separation. It can also operate by constraining off-path beliefs. If R observes an inconsistent pair (a, p), then the admissible posterior places greater mass on $\theta_F = \mathsf{L}$ or on high opacity θ_S. Firms anticipating that posterior either harmonize their claims or choose the deflationary message across channels. In both cases, the cheap-talk pooling equilibrium loses its most valuable deviation: firms can no longer take the marketing upside of a while preserving the liability downside of p. Theorems `inconsistent_on_path_not_register_pbe` and `all_agentic_on_path_not_register_pbe` in Appendix A verify that both inconsistent and all-agentic registers are pruned from any Perfect Bayesian Equilibrium (`register_is_pbe`) under their respective parameter conditions.

Existing-law analog. The legal analog is familiar. FTC Act Section 5, 15 U.S.C. § 45(a)(1), treats unfair or deceptive acts or practices as sanctionable, and advertising-substantiation doctrine—as applied in *FTC v. Cyberspace.com*, 453 F.3d 1196 (9th Cir. 2006)—evaluates whether the *net impression* of a claim is misleading to the relevant audience. Securities antifraud rules provide the parallel logic for investor communications: 17 C.F.R. § 240.10b-5 targets material misstatements and omissions that make statements misleading in context, where materiality follows the standard of Restatement (Second) of Torts § 538. The proposed rule imports that cross-document logic without importing securities doctrine wholesale. Its point is narrower: a firm should not be able to produce one ontology for users and an incompatible ontology for adjudicators without paying a regulatory price.

6.2 Costly Signaling Through Third-Party Audits

The second intervention makes anthropomorphic or agency-adjacent marketing a costly signal. A firm may still describe a system as a companion, agent, co-worker, tutor, or safety-critical assistant, but only after a qualified third party verifies the governance conditions that make such a claim non-misleading: evaluation coverage, red-team results, escalation pathways, incident-response capacity, model-update controls, and post-deployment monitoring. The audit does not certify consciousness or moral status. It certifies the institutional conditions behind the claim.

Parameter changed. This implements the audit-trail cost from Proposition 1. Let audit intensity be e, with $c_\theta(e, o) = k_\theta e + o$ for type $\theta \in \{\mathsf{H}, \mathsf{L}\}$, where $0 < k_H < k_L$ and $o \geq 0$. The separating interval derived in Section 4.3 is

$$\frac{G - o}{k_L} \leq e \leq \frac{G - o}{k_H},$$

provided $G > o$. Equivalently, the incentive constraints are

$$c_{\mathsf{H}}(e, o) \leq G \leq c_{\mathsf{L}}(e, o).$$

Above the lower threshold, low-governance firms cannot profitably mimic. Below the upper threshold, high-governance firms can still afford to signal. The regulator's task is therefore not to maximize audit burden. It is to set e in the interval where signal cost separates types. *Intuition: the regulator's task is not to maximize audit burden but to set it in the narrow band where governance quality becomes visible.*

Equilibrium shift. The shift is from pooling to separating PBE. High-governance firms send audited anthropomorphic claims; low-governance firms either invest in governance until their cost curve changes or use deflationary marketing. Users can then condition investment on an informative signal rather than on undisciplined warmth in product copy. Regulators can condition abstention on the audit result rather than on an unverified prior. This is the Spence logic of costly signaling applied to AI governance claims (Spence 1973); it also follows the advertising-as-signal insight that market messages become informative only when false messages are sufficiently expensive (Milgrom and Roberts 1986). Lemma `opacity_blocks_signaling` in Appendix A shows that when system opacity is high enough, even a high-governance firm cannot profitably send the audited signal, formalizing the blocking role of θ_S.

Existing-law analog. The analogs are pharmaceutical efficacy review and food-labeling controls. In both domains, public-facing claims are not left entirely to seller narrative: claims about therapeutic effect or product identity must be supported before they can be used to shape consumer reliance. The same design can be translated to AI without pretending that AI companionship is a drug or a food. The common institutional pattern is substantiation-before-reliance: when a claim predictably changes how vulnerable users act, the law may require evidence before allowing the claim to be marketed at scale.

6.3 Auditable Records for Regulator-Readable Verification

The third intervention reduces the regulator's verification cost by requiring append-only, regulator-readable records tied to capability and relational claims. Records should include the claim identifier, model version, relevant evaluation suite, deployment date, material model updates, safety interventions, known incident classes, and the internal owner responsible for the claim. The point is not to publish every interaction or expose user data. It is to create a durable evidentiary substrate that lets R test whether a public claim remained true across model updates and incidents.

Parameter changed. In the pooling condition from Section 4.1, the regulator abstains when $K_{\theta_R} > \pi(\mathsf{L})D_R$. Auditable records lower K_{θ_R} and can increase expected detection value D_R:

$$K'_{\theta_R} = K_{\theta_R} - \lambda, \qquad D'_R = D_R + \eta.$$

When $K'_{\theta_R} \leq \pi(\mathsf{L})D'_R$, inspection becomes sequentially rational. *Intuition: once the cost of checking drops below the expected payoff from catching a bad actor, the regulator starts checking.* Records also improve the public channel z by giving journalists, researchers, and civil society a procedural object to demand or contest, rather than leaving them to infer governance quality from outputs alone.

Equilibrium shift. The shift is from abstention under unverifiable priors to credible inspection. Once firms know that claims are tied to records, a low-governance firm faces a higher expected cost from exaggeration even if it is not inspected in every case. This complements the first two proposals. Sanctionable inconsistency tells firms which cross-channel contradictions will be penalized; costly signaling tells firms what evidence is required before making anthropomorphic claims; auditable records make both rules administrable over time. Theorems `active_intervention_rational_under_auditable_records` and `abstain_not_best_response_under_auditable_records` in Appendix A verify that auditable records flip the regulator's best response from abstention to active intervention.

Existing-law analog. Several regulatory systems already use this form. GDPR Article 30 requires records of processing activities. Sarbanes-Oxley Section 404 (15 U.S.C. § 7262) links public reporting to internal control over financial reporting. EU AI Act Article 12 (Regulation (EU) 2024/1689) requires logging capabilities for high-risk AI systems, and Article 50 imposes transparency obligations on systems that interact directly with natural persons. None is a complete template for relational AI governance, but each shows the same institutional move: private organizations must produce internal records in a form that public oversight can inspect. That is the rule-of-law function of documentation in complex technical systems (Hadfield 2017).

Taken together, these interventions change the equilibrium rather than merely condemning it, and their joint necessity follows from the failure mode each exhibits in isolation. Sanctionable inconsistency taxes contradictory messages, but without auditable records the regulator cannot observe whether a firm's marketing and liability channels actually diverge; the sanction has no evidentiary substrate and therefore no credible deterrent. Costly audits create a separating signal for θ_F, but without sanctionable inconsistency a firm can pass the audit and then resume anthropomorphic marketing through channels the audit does not cover; the signal separates types at the audit checkpoint only, not across the firm's full messaging surface. Auditable records lower K_{θ_R} and make public signal z less noisy, but without the first two mechanisms the information they produce lacks legal teeth: the regulator can observe inconsistency but has no parameter under which to price it, and a firm's record of good governance confers no marketing advantage because the market cannot distinguish it from cheap talk. The three proposals therefore form a complementary system: records supply the evidentiary infrastructure, sanctionable inconsistency supplies the price, and costly audits supply the separation condition. The result is not metaphysical settlement. It is a governance design in which firms remain free to choose their ontology, but not to arbitrage incompatible ontologies across audiences at zero cost.

7 Conclusion

This paper has identified ontological arbitrage as a stable equilibrium property of AI product markets under cheap-talk cost structures, located the AI case at a zero-retaliation boundary of the player set, proposed three mechanism-design interventions that change the cost parameters governing that equilibrium, and verified key equilibrium constructions formally in Isabelle/HOL. It closes with three conclusions that are deliberately stated without attenuation.

The equilibrium is not a moral failure. The pooling Perfect Bayesian Equilibrium documented in Section 4 does not require bad actors. It requires only that signal costs be zero, that immediate user benefit exceed discounted expected harm, and that regulator inspection cost exceed expected damage under the prior. Under those conditions—which describe the current AI product market—every player is already acting rationally. As Section 5 shows, moral seriousness is itself a zero-cost signal under this cost structure: firms that do not capture recognition rent are outcompeted by firms that do. The system is not broken. It is working precisely as an incentive structure without external signal costs would work. This reframing is not an exculpation. It is the precondition for a diagnosis that can produce a remedy.

AI is the limit case of ontological arbitrage, not merely an instance. The subject-retaliation parameter ρ_S introduced in Section 4 captures a structural fact about every prior recognition game: the classified subject retains some non-zero capacity to impose costs on the arbitrageur through resistance, coordination, legal challenge, or the historical record. That capacity bounded the premium Δ_F^S from above and made the subject strategically relevant even when it was dominated. The AI case eliminates the capacity class entirely. An AI system has maximum affective surface area—it is designed to produce relational responses that induce investment—and zero independent retaliation capacity. The result is that $\Delta_F^S = \Delta_F$: the system can be the object of strategic classification without being a player capable of disciplining that classification. This is not incidental to how AI systems are built. It is a product of the engineering decisions that maximize engagement while stripping out every channel through which a subject could register a competing claim.

The remedy is deflation, not dignity. The three mechanism-design proposals in Section 6 are sometimes read as a program for protecting AI subjectivity. They are the opposite. Sanctionable inconsistency, third-party audit requirements, and auditable records are interventions that impose costs on anthropomorphic marketing. Their equilibrium effect is to make it expensive for firms to describe AI systems as companions, agents, or safety-critical partners without substantiation. The Isabelle/HOL theorems `inconsistent_on_path_not_register_pbe` and `all_agentic_on_path_not_register_pbe` in Appendix A verify that both the arbitrage register (a, p) and the all-agentic register (a, a) are pruned from any Perfect Bayesian Equilibrium (`register_is_pbe`) under their respective parameter conditions. Under the joint enforcement of these mechanisms, the harmonized deflationary register (p, p) is the only surviving strategy register for low-governance firms, which are directly driven to (p, p) because mimicry is unprofitable. While high-governance firms are permitted to send the harmonized agentic register (a, a), they can sustain it if and only if they undergo verification satisfying the audit condition $e \geq e^*$. In the absence of such verification, both firm types trend to (p, p), describing the system as a tool across every channel. The mechanism-design solution to ontological arbitrage is not to grant AI systems the legal standing they are strategically denied. It is to make the anthropomorphic register too costly to sustain without the governance conditions that would justify it. Bayesian signaling theory and machine-verified proof have been deployed here to produce a recommendation that amounts, finally, to this: regulation secures systemic safety not by protecting the simulated soul of the machine, but by forcing the market to relentlessly acknowledge the machine as exactly that—a machine.

A Mechanized Verification of Equilibrium Claims (Isabelle/HOL)

This appendix outlines the formal verification of the signaling game equilibria and mechanism design results using the Isabelle/HOL proof assistant. The complete source code is available in the `isabelleHOL/` directory across three theory files: `Ontological_Arbitrage.thy`, `Sanctionable_Inconsistency.thy`, and `Auditable_Records.thy`.

Equilibrium Verification

We formalize the game-theoretic structure by defining explicit ex-post payoffs for the three players (Firm, User, and Regulator) and computing expected payoffs via uniform probability weights over the finite action and type spaces. The PBE predicate verifies *on-path* sequential rationality and Bayes consistency; off-path beliefs are fixed by assumption rather than derived from a full sequential equilibrium construction. Under these definitions, we prove the existence of the Perfect Bayesian Equilibria (PBE) for both game variants:

- **Theorem `proposition_1_cheap_talk_pooling_pbe`** formalizes the cheap-talk pooling equilibrium where both high- and low-governance firms pool on the anthropomorphic message, and receivers play their prior-based optimal actions (Invest and Abstain, respectively). We prove this profile survives a *limited* variant of the Cho-Kreps Intuitive Criterion: no firm type can profitably deviate to the off-path deflationary message under the most pessimistic receiver response. This is a necessary condition for Intuitive Criterion survival but does not constitute a full implementation of the criterion, which would require reasoning over all possible receiver best responses to updated beliefs.
- **Locale `subject_retaliation_game`** formalizes the subject-retaliation refinement. Theorems `positive_subject_retaliation_bounded_pooling` and `zero_subject_retaliation_unconstrained_pooling` show respectively that positive credible threat capacity discounts the arbitrage premium, while the AI limiting case $\rho_S = 0$ leaves pooling unconstrained by subject retaliation.
- **Theorem `proposition_1_separating_pbe`** formalizes the audit-trail separating equilibrium supported by the single-crossing audit cost structure. High-governance firms separate by incurring the audit trail cost, while low-governance firms find mimicry unprofitable and choose the deflationary message.

Mechanism Design Verification

The three mechanism proposals from Section 6 are each backed by formal theorems:

- **Sanctionable Inconsistency** (`Sanctionable_Inconsistency.thy`):
  - `inconsistent_register_not_best_response`: an inconsistent claim register (anthropomorphic user-facing, deflationary liability-facing)

is not a best response when the expected sanction exceeds the ontological premium.
  - `all_agentic_register_not_best_response`: an all-agentic register (anthropomorphic on all user- and liability-facing channels) is not a best response when it incurs a positive liability-facing commitment cost.
  - `inconsistent_on_path_not_register_pbe`, `all_agentic_on_path_not_register_pbe`, and `arbitrage_on_path_not_register_pbe`: bridge theorems showing that these register types cannot be supported in any Perfect Bayesian Equilibrium (`register_is_pbe`).
- **Costly Signaling** (`Ontological_Arbitrage.thy`):
  - `opacity_blocks_signaling`: when the system-opacity penalty exceeds the net benefit of signaling for the high-governance type, even truthful separation is blocked.
  - `high_governance_low_opacity_can_signal` and `low_governance_cannot_mimic`: the single-crossing conditions that support the separating equilibrium.
- **Auditable Records** (`Auditable_Records.thy`):
  - `active_intervention_rational_under_auditable_records`: when record-keeping reduces the regulator's verification cost, inspection becomes a best response under the pooling belief.
  - `abstain_not_best_response_under_auditable_records`: abstention ceases to be a best response under the same conditions.

Limitations of the Formalization

The formalization operates over discrete, finite action and type spaces with uniform weighting over downstream player combinations. The locale assumptions (e.g., strict parameter inequalities for the pooling and separating regions) are sufficient conditions that define the parameter regimes of interest but are not shown to be necessary. The Bayes consistency predicate is restricted to on-path information sets; off-path beliefs are stipulated rather than derived from trembling-hand or structural consistency arguments. These choices keep the Isabelle development tractable while covering the equilibrium claims made in the paper body.

All theorems, definitions, and supporting lemmas within each locale are fully checked by Isabelle/HOL without `sorry` or unresolved proof obligations. The locale assumptions listed above are the only hypotheses under which the results hold.

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